

## 参考文献

- [1] PCISIG. PCI local bus specification, revision 3.0 [S]. 2003.
- [2] PCISIG. PCI-to-PCI bridge architecture specification, revision 1.2 [S]. 2003.
- [3] PCISIG. PCI express base specification, revision 2.1 [S]. 2009.
- [4] PCISIG. PCI express base specification, revision 3.0, version 0.7 [S]. 2009.
- [5] PCISIG. PCI express to PCI/PCI-X bridge specification, revision 1.0 [S]. 2003.
- [6] PCISIG. PCI bus power management interface specification, revision 1.2 [S]. 2004.
- [7] PCISIG. address translation services, revision 1.1 [S]. 2009.
- [8] PCISIG. Single root I/O virtualization and sharing specification, revision 1.0 [S]. 2007.
- [9] PCISIG. Multi-root I/O virtualization and sharing specification, revision 1.0 [S]. 2008.
- [10] SHANLEY T, ANDERSON D. “Chapter 25: Transaction Ordering & Deadlocks”, PCI system architecture [M]. 4th ed. Reading: Addison Wesley, 1999: 651-671.
- [11] BUDRUK R, ANDERSON D, SHANLEY T. PCI express system architecture, chapter 5 ACK/NAK protocol [M]. Reading: Addison Wesley, 2003.
- [12] Intel. 21555 Non-Transparent PCI-to-PCI bridge user manual [S]. 2001.
- [13] Intel. Intel 64 and IA-32 architectures software developer’s manual volume 3A: system programming guide, part 1 [S]. 2008.
- [14] Intel. Intel 64 and IA-32 architectures software developer’s manual volume 3B: system programming guide, part 2 [S]. 2008.
- [15] Intel. mobile intel® 4 series express chipset family, revision 2.1 [S]. 2008.
- [16] Intel. Intel I/O controller hub 9(ICH9) family datasheet [S]. 2008.
- [17] Intel. Intel virtualization technology for directed I/O [S]. 2008.
- [18] Intel. Interrupt swizzling solution for intel 5000 chipset series-based platforms [OL]. [http:// www.intel.com/assets/pdf/appnote/314337.pdf](http://www.intel.com/assets/pdf/appnote/314337.pdf). 2006.
- [19] Intel. Multiprocessor specification version 1.4 [S]. 1997.
- [20] Intel. ACPI component architecture programmer reference—OS-independent subsystem, debugger, and utilities, Revision 1.22 [S]. 2009.
- [21] IBM. Power ISA, version 2.05[S]. 2007.
- [22] Freescale. PowerPC e500 core family reference manual, revision 1 [S]. 2005.
- [23] Freescale. MPC8548E PowerQUICC™ III integrated processor family reference manual, revision 2 [S]. 2007.
- [24] Freescale. MPC8572E PowerQUICC™ III integrated processor family reference manual, revision A [S]. 2008.
- [25] Freescale. QorIQ P4080 communications process product brief [S]. 2009.
- [26] Freescale. Embedded multicore: an introduction [S]. 2009.
- [27] AMD. AMD64 technology—AMD64 architecture programmer’s manual volume 2: system programming [S]. 2007.
- [28] AMD. AMD I/O virtualization technology (IOMMU) specification [S]. 2009.
- [29] Xilinx. LogiCORE™ Endpoint PIPE v1.7 for PCI Express® User Guide [S]. 2007.
- [30] HP, Intel, Microsoft, Phoenix, Toshiba. Advanced configuration and power interface specification 4.0

- [S]. 2009.
- [31] MAPDOX R A, SINGH G, SAFRANEK R J. Weaving high performance multiprocessor fabric [M]. Santa Clara: Intel Press, 2009.
- [32] GARBUS E, SANKHAGOWIT P, GODSCHMIDT M, et al. Architecture for an I/O processor that integrates a PCI to PCI bridge [P]. 1999.
- [33] WINKLES J. Sizing of the replay buffer in PCI express devices [OL]. [http://www.mindshare.com/files/resources/MindShare\\_PCIe\\_Elastic\\_Buffer.pdf](http://www.mindshare.com/files/resources/MindShare_PCIe_Elastic_Buffer.pdf). MindShare, Inc. 2003.
- [34] WINKLES J. Elastic buffer implementations in PCI express devices [OL]. [http://www.mindshare.com/files/resources/MindShare\\_PCIe\\_Elastic\\_Buffer.pdf](http://www.mindshare.com/files/resources/MindShare_PCIe_Elastic_Buffer.pdf). MindShare, Inc. 2003.
- [35] TIPLEY R E. Split transaction protocol for the peripheral component interconnect bus;5533204 [P]. 1996-07-02.
- [36] James E Smith. A Study of Branch Prediction Strategies, Proceedings of the 8th Annual Symposium on Computer Architecture [J]. p.135-148, May 12-14, 1981, Minneapolis, Minnesota, United States. 1981.
- [37] T Y Yeh, Y N Patt. Alternative Implementation of Two-level Adaptive Branch Prediction [J]. Computer architecture, 1992; 124-134.
- [38] Intel. The white paper of Intel® next generation nehalem microarchitecture [OL]. <http://www.intel.com/technology/architecture-silicon/next-gen>. 2009.
- [39] VANDERWIEL S P, LILJA D J. Data prefetch mechanisms [J]. ACM computing surveys, 2002, 32(2): 26-38.
- [40] JOUPPI N P. Improving direct-mapped cache performance by the addition of a small fully-associative cache and prefetch buffers [J]. Proceedings of the 17th annual international symposium on computer architecture, 1990; 364-373.
- [41] LIPASTI M H, WILKERSON C B, SHEN J P. Value locality and load value prediction [J]. Proceedings of the seventh international conference on architectural support for programming Languages and operating systems, 1996; 138-147.
- [42] BONOMI F, FENDICK K W. The rate-based flow control framework for available bit-rate ATM services [J]. IEEE network, 1995; 25-39.
- [43] DALLY W J. Virtual-channel flow control [J]. IEEE Transactions On Parallel and Distributed System, 1992, 3(2): 13-17.
- [44] H T Kung, Robert Morris. Credit-Based Flow Control for ATM Networks [J]. IEEE Network, 1995; 40-48.
- [45] KUNG H T, BLACKWELL T, CHAPMAN A. Credit-based flow control for ATM networks: credit update protocol, adaptive credit allocation, and statistical multiplexing [J]. Proceedings of the ACM SIGCOMM 1994 symposium on communications architectures, protocols, and applications, 1994; 101-104.
- [46] KUANG H T, CHAPMAN A. The FCVC(Flow-controlled Virtual Channels) Proposal for ATM Networks [S]. 1993.
- [47] PAREKH A, GALLAGER R. A generalized processor sharing approach to flow control: the single node case [J]. In technical report LIDS-TR-2040, laboratory for information and decision systems, massachusetts institute of technology, 1991.
- [48] PAREKH A. A generalized processor sharing approach to flow control in integrated services networks [J]. In Technical Report LIDS-TR-2089, Laboratory for Information and Decision Systems, Massachusetts Institute of Technology, 1992.
- [49] KERNAMI P, KLEINROCK L. Virtual cut through: a new computer communication controller [J]. IEEE Computer Society Press, 1987; 230-234.

- [50] SHREEDHAR M, VARGHESE G. Efficient fair queueing using deficit round-robin [J]. IEEE/ACM Transactions on Networking, 1996, 4(3): 24-32.
- [51] FERGUSON P, HUSTON G. Quality of service: delivering QoS on the internet and in corporate networks [M]. Hoboken: John Wiley & Sons, Inc., 1998.
- [52] DALLY W J. Performance analysis of k-ary n-cube interconnection networks [J]. IEEE Transactions on Computers, 1990, 39(6): 775-785.
- [53] REGULA J. Using PCIe in a variety of multiprocessor system configurations [OL]. [http:// www.plxtech.com/about/news/archive/articles2007](http://www.plxtech.com/about/news/archive/articles2007). PLX Technology. 2007.
- [54] ALBERT X, FRANASZAK A. A DC-balanced, partitioned-block, 8B/10B transmission code [J]. IBM J. RES. DEVELOP, 1983, 27(5): 362-376.
- [55] BOTTOMLEY J E J. Dynamic DMA mapping using the generic device [OL]. <http://www.kernel.org/doc/Documentation/DMA-API.txt>.
- [56] HANDY J. Cache memory book [M]. 2nd ed. Burlington: Morgan Kaufmann, 1998.
- [57] KRISHNAN V. Towards an integrated IO and clustering solution using PCI express [J]. IEEE International Conference on Cluster Computing (CLUSTER 2007), 2007.
- [58] HUM H, GOODMAN J A. Forward state for use in cache coherency in a multiprocessor system; US6922756 [P]. 2002-12-19.
- [59] REGULA J. Using PCIe in a variety of multiprocessor system configurations[OL]. [http:// www.embedded.com/columns/technicalinsights/196902357?\\_requestid=349156](http://www.embedded.com/columns/technicalinsights/196902357?_requestid=349156). 2007.
- [60] HOLDEN B. Latency comparison between hyperTransport and PCI-express in communication systems[OL]. [http:// enterprise2. and. com/Downloads/Industry/Telecommunications/Latency \\_ Comparison \\_ Hyper-Transport. pdf](http://enterprise2.and.com/Downloads/Industry/Telecommunications/Latency_Comparison_Hyper-Transport.pdf). 2006.
- [61] GOLDHAMMER A, JR J A. Understanding performance of PCI express system [OL]. [http://www.xilinx.com/support/documentation/white\\_papers/wp350. pdf](http://www.xilinx.com/support/documentation/white_papers/wp350.pdf). 2008.
- [62] SBS Implementers Forum. System management bus specification version 2.0 [S]. 2000.
- [63] Application Notes from Maxim. Comparing the I<sup>2</sup>C bus to the SMBus [OL]. [http://pdfserv.maxim-ic.com/en/an/AN476. pdf](http://pdfserv.maxim-ic.com/en/an/AN476.pdf). 2000.
- [64] PLX. ExpressLane PEX 8518AA/AB/AC 5-Port/16-Lane PCI Express Switch Data Book [S]. 2007.